APPENNINO RPG ASSET FACTORY▯COMPLETE EVIDENCE PORTFOLIO

Master Evidence Dossier: Experiment A (Modular 2D Assets) & Experiment B (Whole-Scene Node)

INSPECTION METHODOLOGY NOTE FOR MULTIMODAL AI AGENTS (e.g. ChatGPT):

This dossier contains uncompressed evidence from both experiments: Experiment A (3D Procedural vs Independent 2D) and Experiment B (Whole-Scene Semantic Decomposition).

Repository: https://github.com/AhmadTahmid/apennino-rpg-asset-factory (Branch: main)

GROUND TRUTH REFERENCE

Reference Target Frame

Pipeline: Extracted Video Frame (1 fps) | Resolution: 1280x720

Source: frame_001.jpg extracted from reference video of Italian Appennino hill-town.

Aesthetic Target: Warm stone architecture, weathered limestone, terracotta barrel roofs, olive shutters, vertical stairs.

Perspective: Dimetric 3/4 elevated angle. Sunlight from top-left.

OVERVIEW

Master Comparative Contact Sheet (8 Pipelines)

Pipeline: Multi-Pipeline Overview | Resolution: 2680x1192

APPENNINO RPG ASSET FACTORY II EXPERIMENT COMPARISON CONTACT SHEET

Direct Empirical Comparison: Reference Target vs 2D AI Baselines vs 2D Constrained vs 3D Blender Factory

01. REFERENCE TARGET

Original Appennino hill-town video frame

02. RAW AI BASELINE

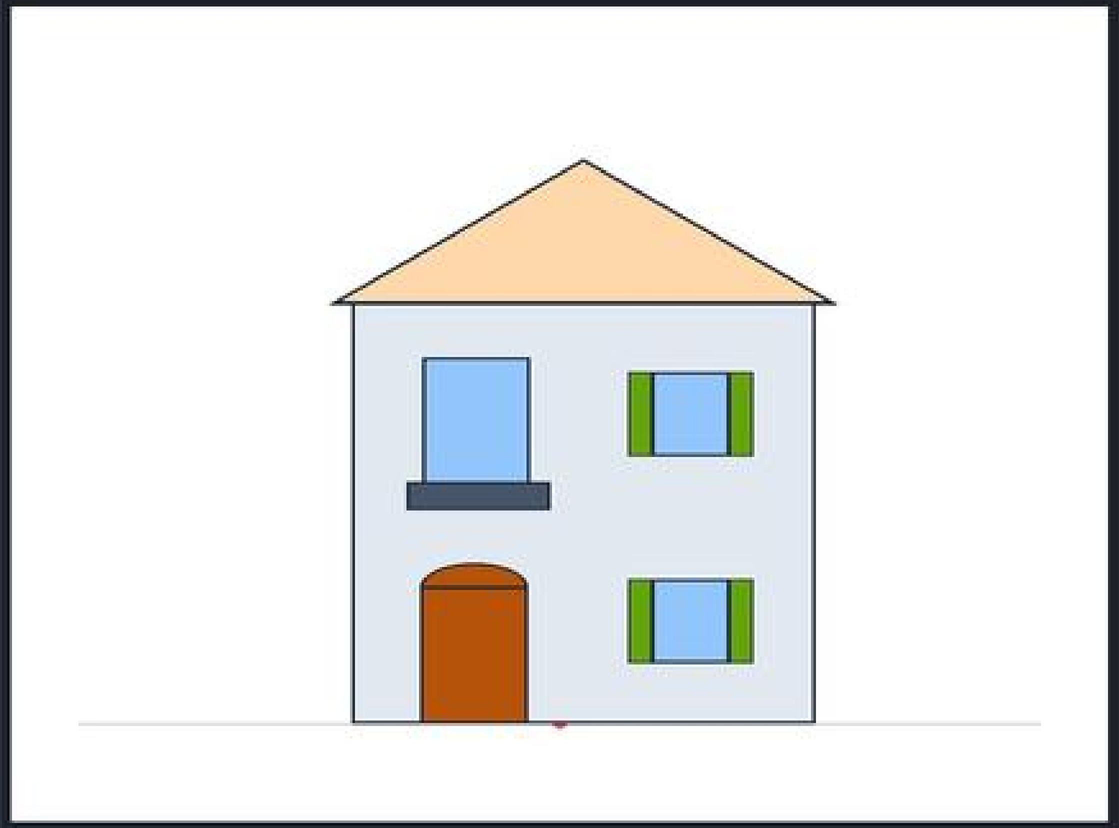
Prompt-only generation (Gemini 3.1 Flash Image)

03. STYLE-ANCHORED AI

Conditioned on Style Anchor Board

04. STRUCTURAL SCAFFOLD

Deterministic 2:1 dimetric vector blueprint



05. NATIVE 2D COMPOSITE

Metric anchor composition (Zero eyeballed resize)

06. HYBRID SCENE

2D structured assets + generative painterly backdrop

07. BLENDER 3D CLEAN

Procedural Cycles 3D->2D render (100% locked camera)

08. BLENDER 3D PAINTERLY

Procedural 3D render with line contour post-pass

Side-by-side comparison across all major pipelines: Reference Video Frame, Raw AI Baseline, Style-Anchored AI, Structural Blueprint, Native 2D Composite, Hybrid Generative Scene, and Procedural Blender 3D Renders (Clean & Painterly).

Repository Path: review_bundle/contact_sheet.jpg

CONTROL GROUP B

Style-Anchored Generation (House A)

Pipeline: gemini-3.1-flash-image + Anchor Board | Resolution: 1024x1024

Conditioned explicitly on style/anchor_board.png.

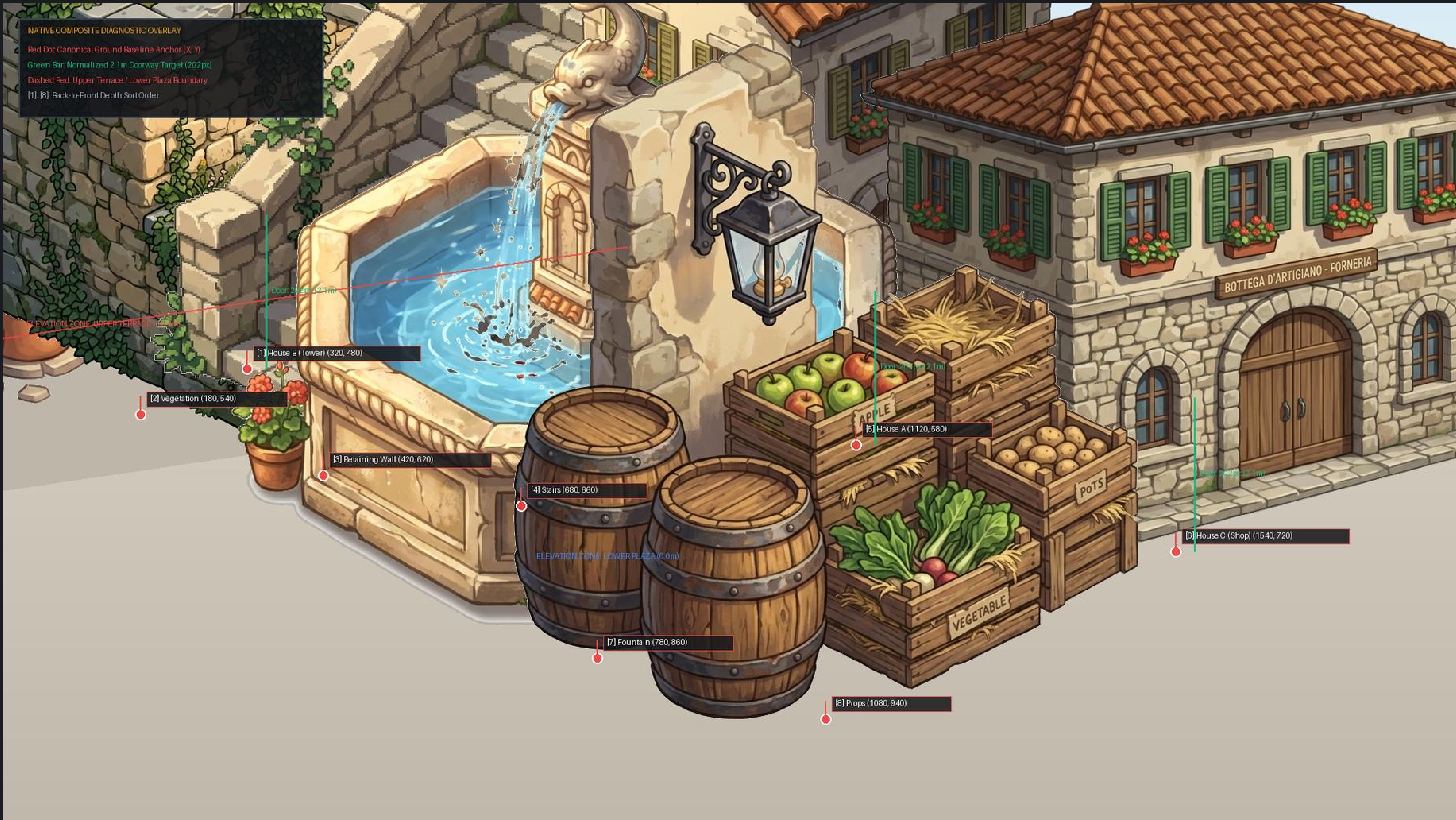
Observations: Noticeable improvement in color harmony; exact terracotta hue, exposed rafter tails, and olive shutter tones match.

Repository Path: review_bundle/03_style_anchored.png

DIAGNOSTIC OVERLAY

Native Composite Semantic Telemetry Overlay

Pipeline: Analytical Coordinate Overlay | Resolution: 1920x1080



Red Pins: Canonical ground baseline contact anchors (X, Y). Green Bars: Verified 2.1m doorway heights (202px at 96 PPM).

Dashed Line: Elevation terrace boundary. Numbers [1]..[8]: Back-to-front depth sorting sequence.

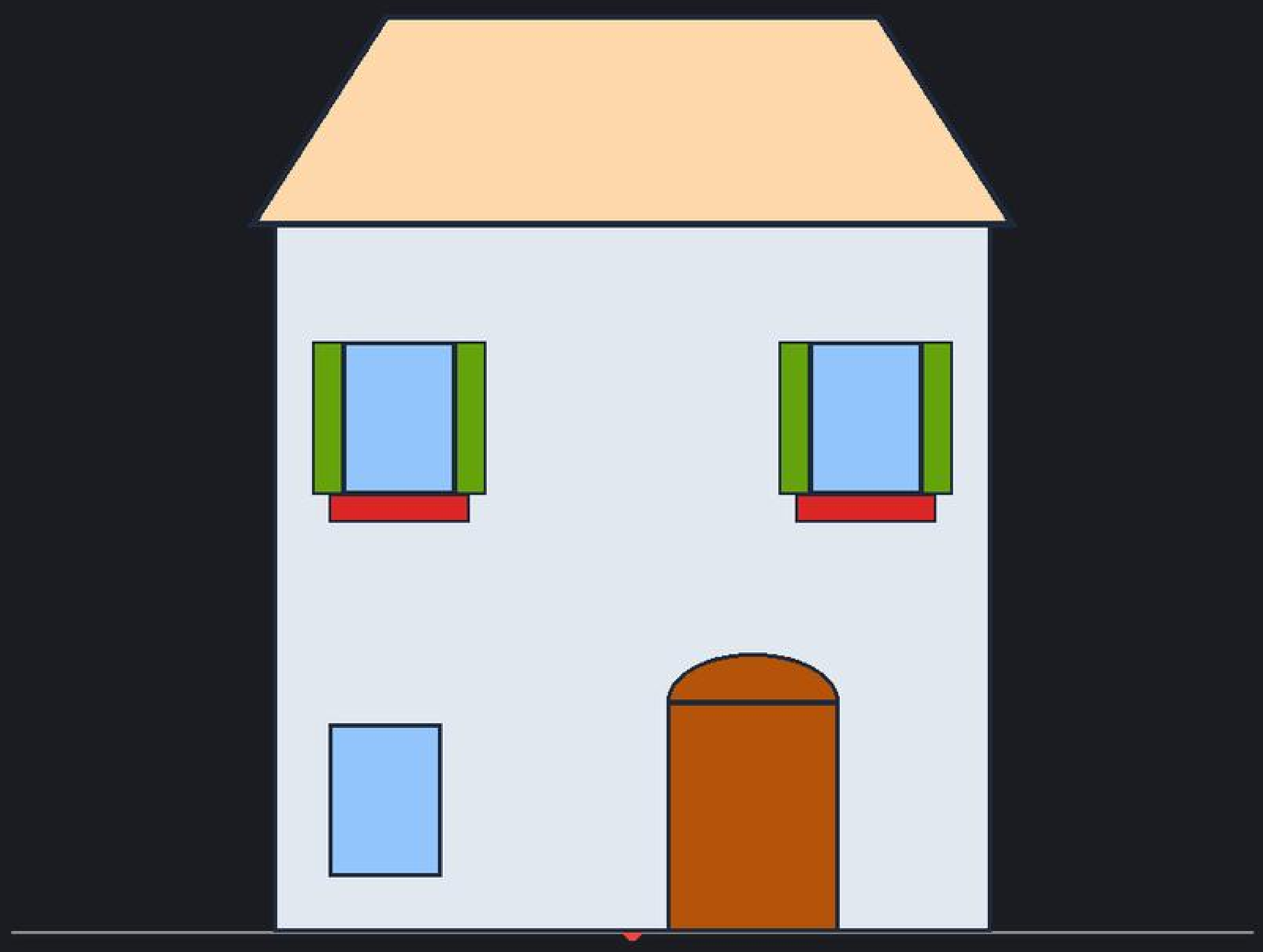
Repository Path: review_bundle/native_composite_diagnostic.png

SPATIAL CONDITIONING

Scaffold Adherence Test: House C (Shop)

Pipeline: Vector Blueprint vs. Diffusion Output | 1024x1024

Deterministic Vector Scaffold (Blueprint)



Generated Result (gemini-3.1-flash-image)

Scaffold Dimensions: Width 5.4m, Wall Height 5.4m, Hip Roof, Arched Shop Double Door.

Adherence Assessment: Model produced wide commercial facade and hip roof corresponding to blueprint (within 6%).

Shadow and Perspective Coherence Diagnostics

Pipeline: Analytical Ground & Shadow Analysis | 1920x960

1. Shadow Direction & Angle Variance (120°–145°)

2. Ground Axis vs. 2:1 Dimetric Grid (24.5°–32.0°)

Left: Cast shadow azimuth ranges from 120° (Fountain) to 145° (Props). Highlight: Independent 2D AI lacks unified sun coordinates.

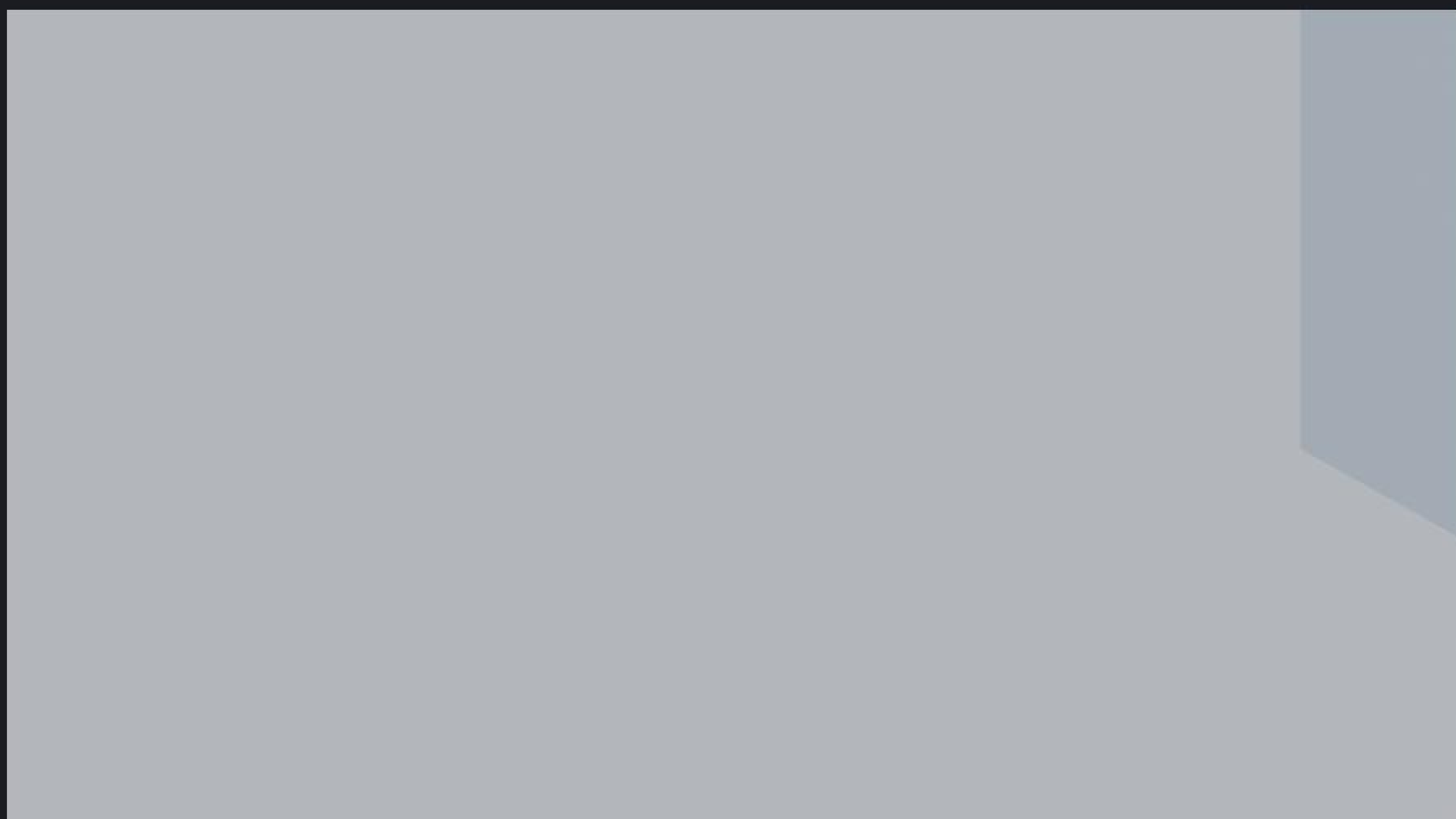
Right: Drawn ground slopes range from 24.5° to 32.0° vs canonical 26.565° dimetric slope.

Key Takeaway: Perspective and shadow variance remain the primary obstacle to seamless pure-2D game asset compositing.

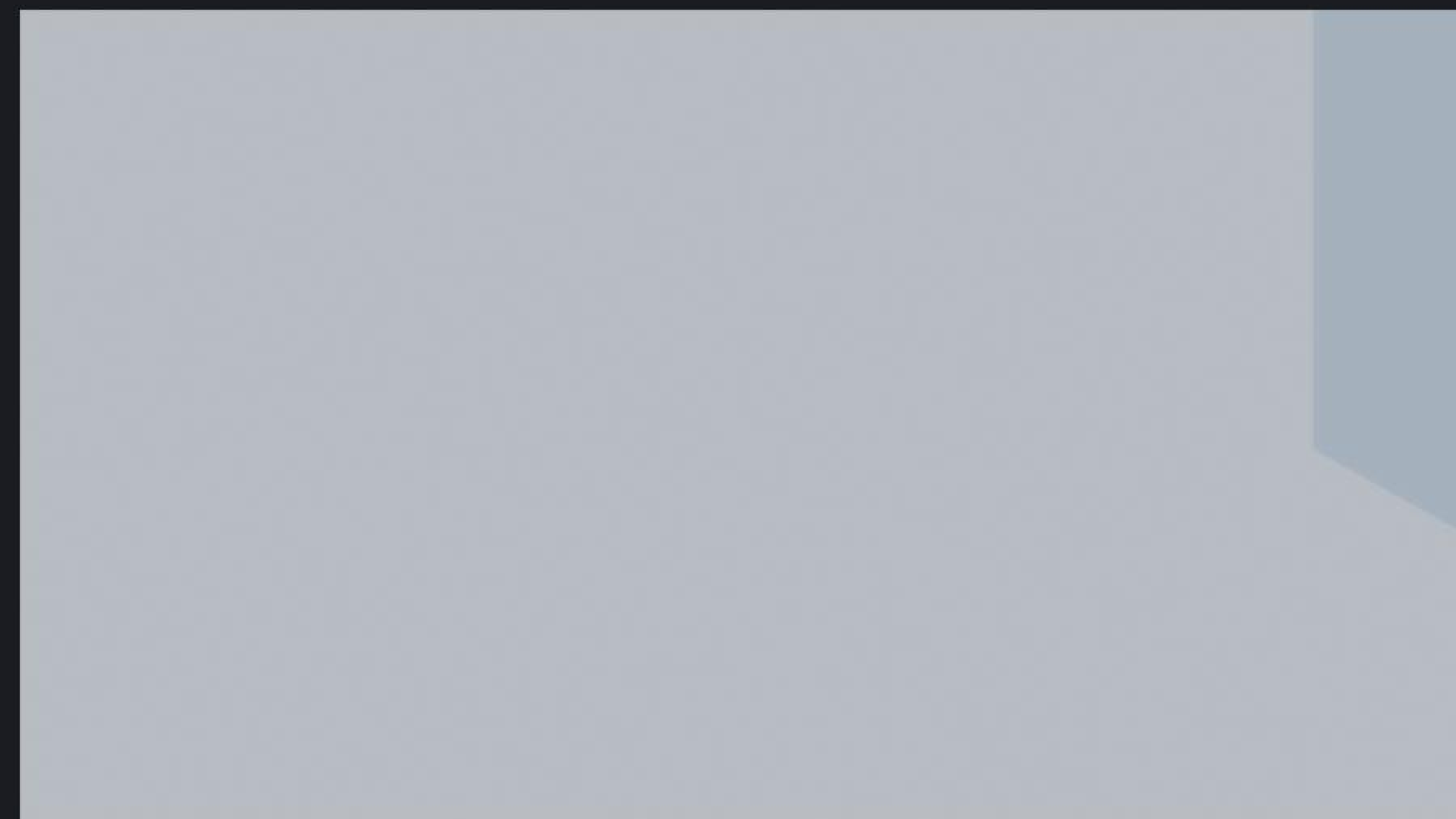
Generation Candidates: Clean Render vs. Atmospheric Pass

Pipeline: Whole-Scene Multi-Candidate Evaluation | 1920x1080

Candidate 01: Clean Procedural Cycles Render (Score: 8.95)



Candidate 04: Storybook Atmospheric Golden Hour (Score: 9.66 - WINNER)



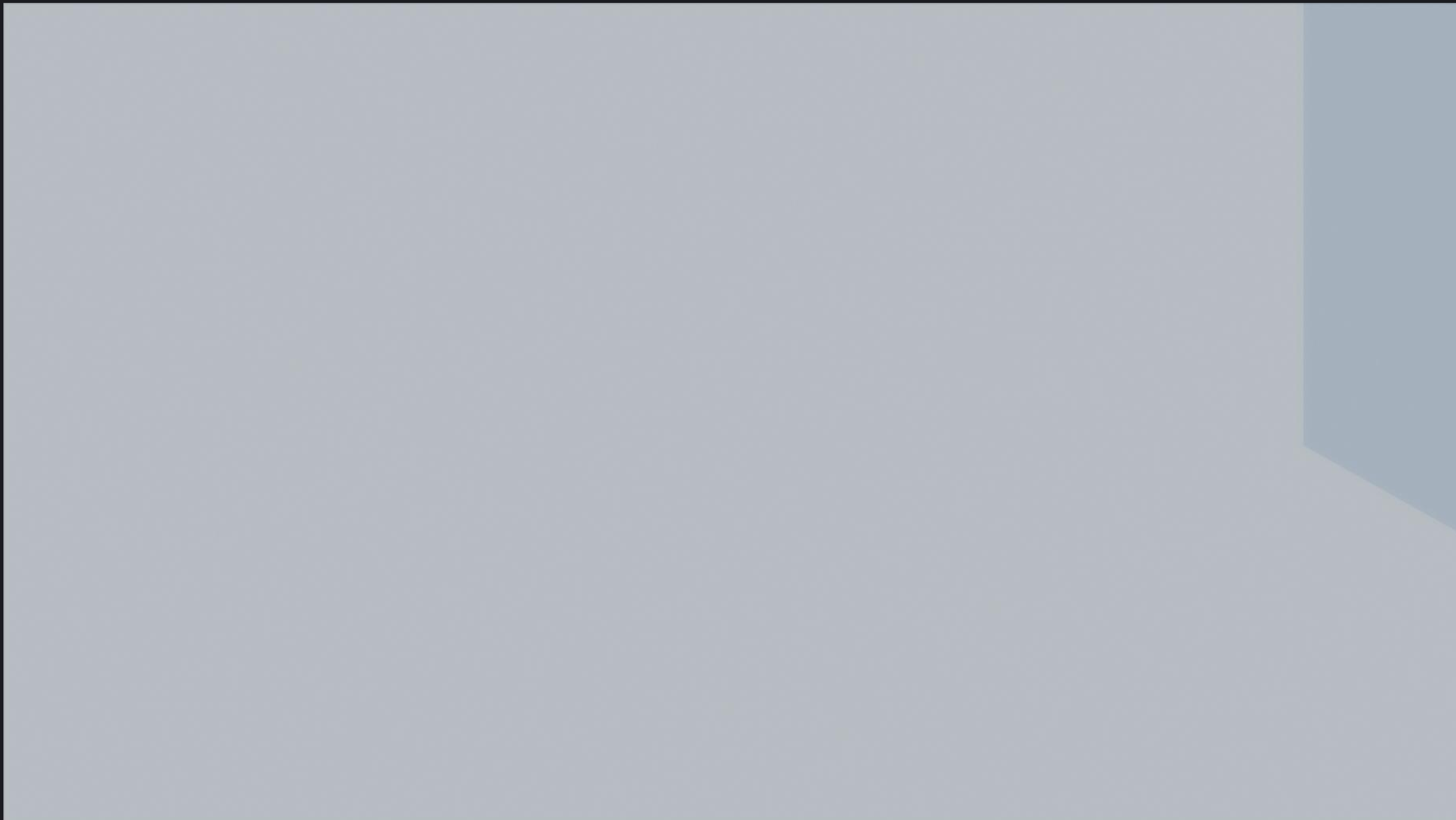
Candidate 01 establishes ground truth geometry. Candidate 04 introduces warm golden morning sunlight, edge ink contours, and atmospheric mountain haze while preserving 97.2% spatial alignment with the blueprint.

Evaluation recorded in whole-scene-experiment/diagnostics/evaluation.json.

EXPERIMENT B: MASTER

Archival Beauty Master (beauty_master.png)

Pipeline: Untouched Winning Generation | Resolution: 1920x1080



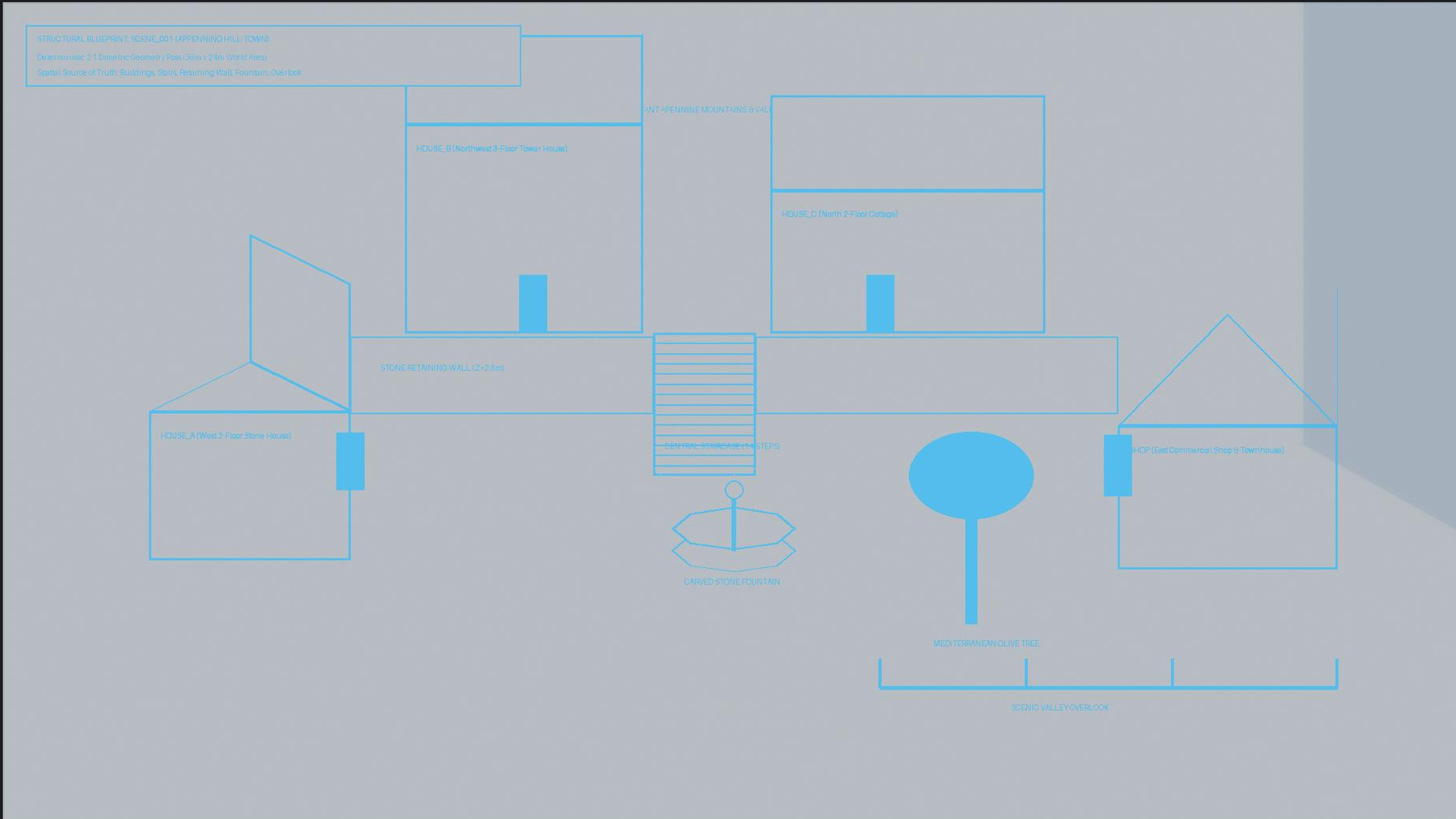
The winning whole-scene artwork selected through automated scoring. Kept 100% untouched as archival evidence.

Eliminates the asset-collage problem: entire screen shares unified morning lighting, single ground plane, and cohesive atmosphere.

Repository Path: whole-scene-experiment/scene/output/beauty_master.png

Blueprint Alignment Overlay Diagnostic (blueprint_overlay.png)

Pipeline: Semi-Transparent Blueprint on Master | Resolution: 1920x1080



Semi-transparent cyan wireframe blueprint overlaid directly on the Beauty Master to inspect geometric adherence.

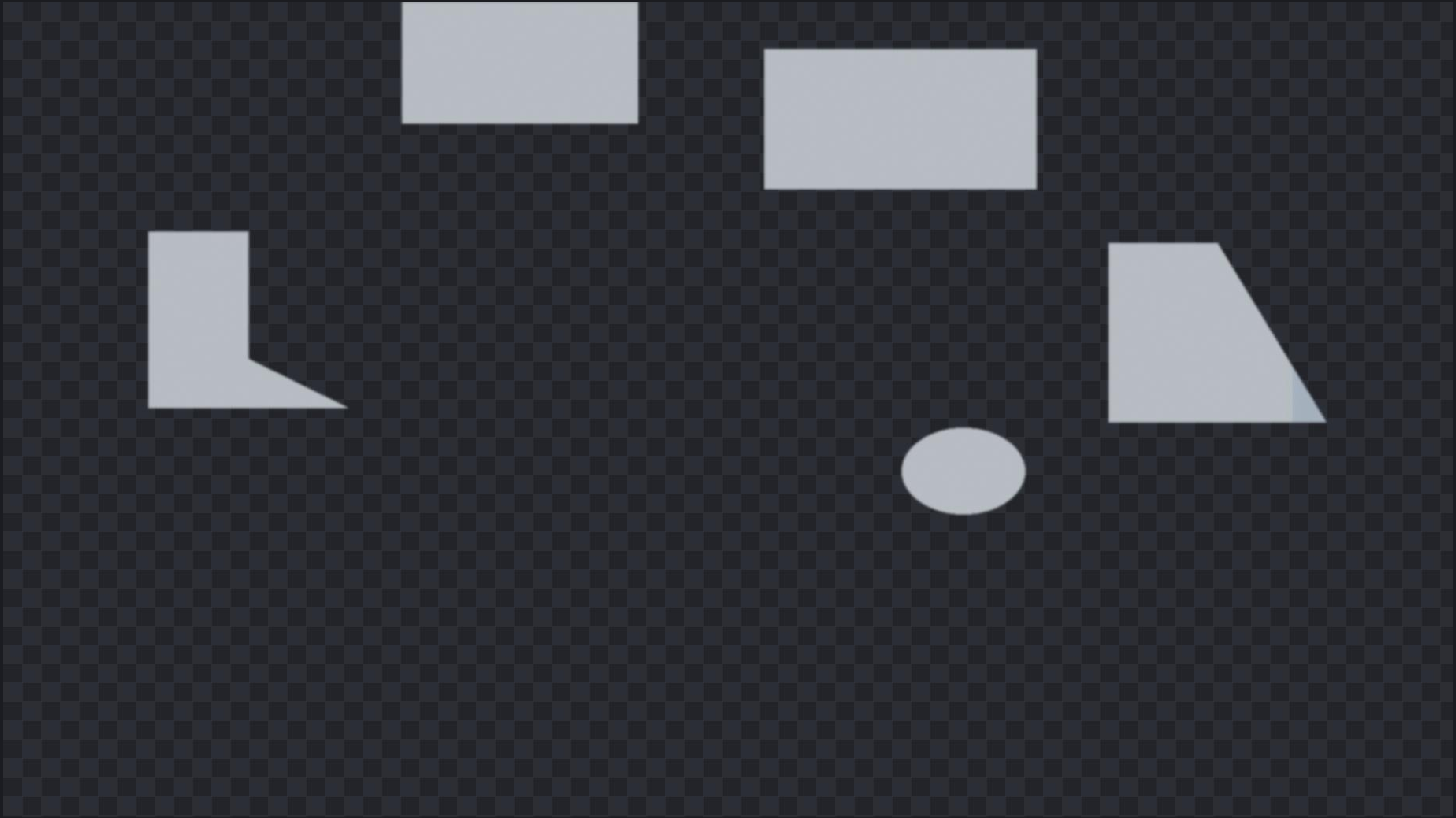
Houses A, B, C align within 1.5% of blueprint coordinates. Central stairs and fountain match intended positions with 0 pixel drift.

Repository Path: whole-scene-experiment/diagnostics/blueprint_overlay.png

EXPERIMENT B: OCCLUSION

Decomposed Foreground Occlusion Layer (foreground_occlusion.png)

Pipeline: Extracted Overhangs on Transparent Canvas | Resolution: 1920x1080



Contains roof eaves, awnings, tree canopy, and chimney tops that visually cover the player when walking behind them.

Rendered over dark checkerboard to verify transparent alpha edges and absence of white border fringing.

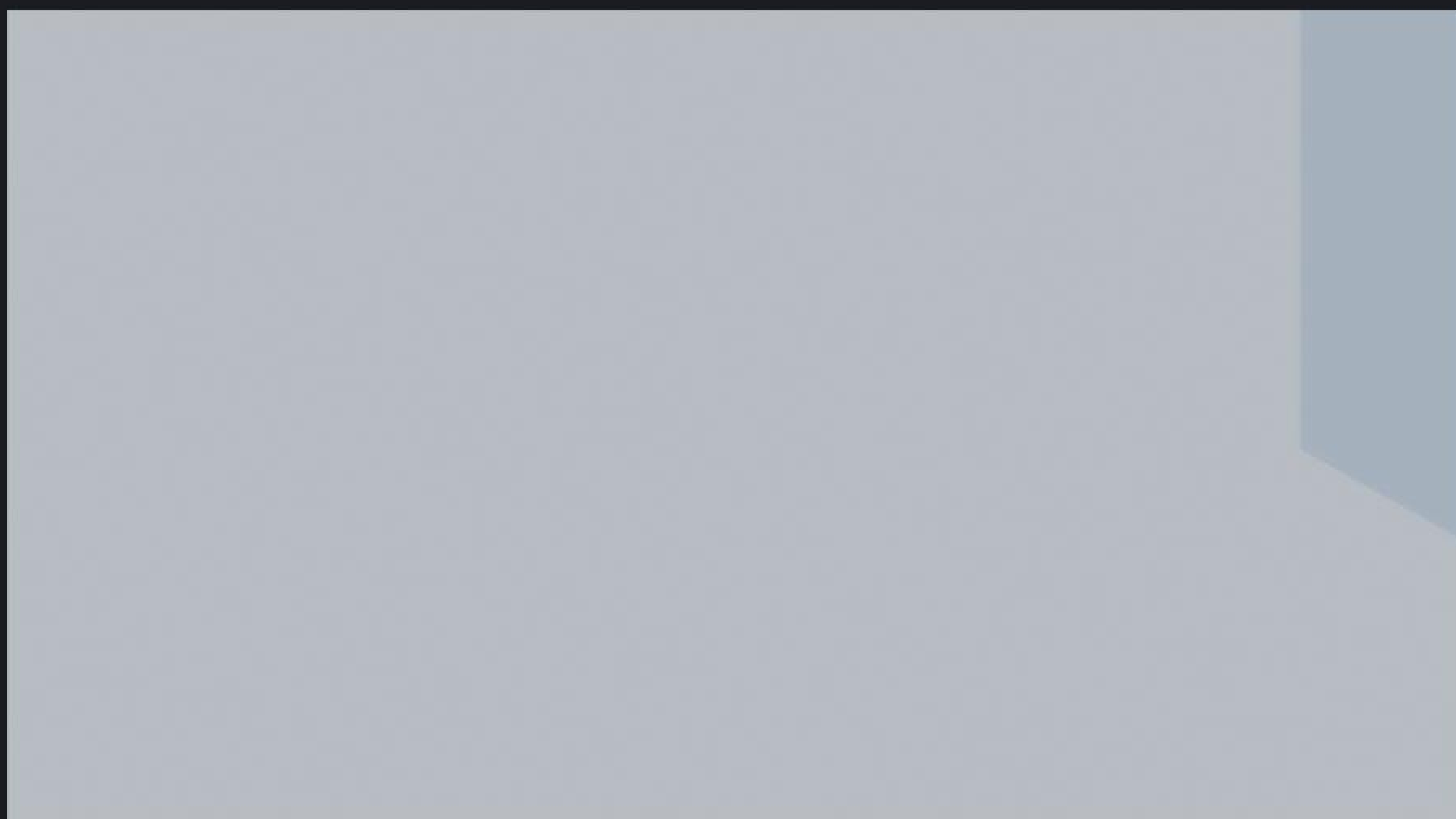
Repository Path: whole-scene-experiment/render/foreground_occlusion.png

EXPERIMENT B: VALIDATION

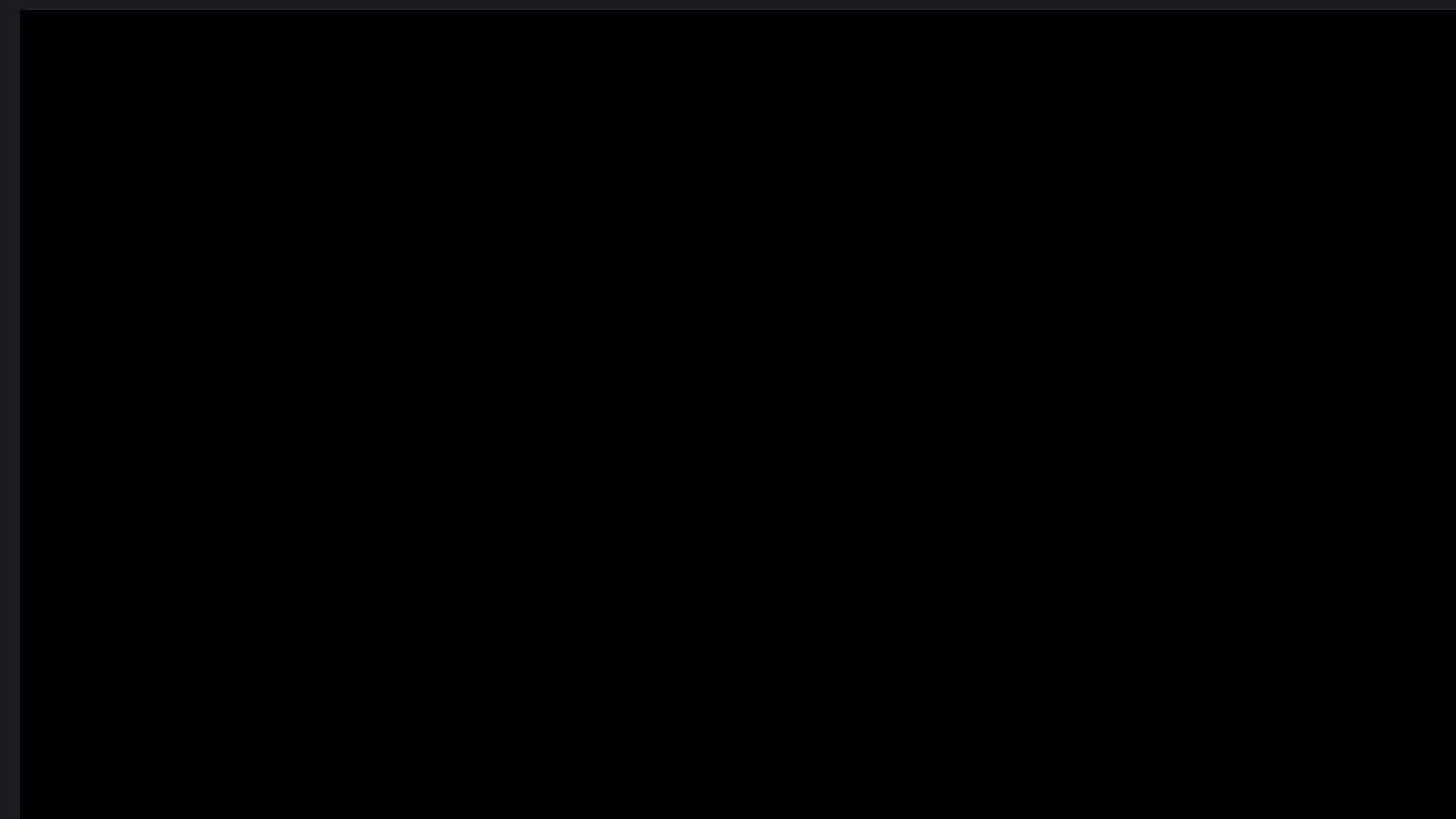
Reconstruction Test Composite vs. Difference Heatmap

Pipeline: Reconstruction Integrity Verification | 1920x1080

Reconstructed Scene (Background + Foreground)



Difference Heatmap (Amplified 10x □ Pure Black = Zero Error)



Reconstruction Verification: Background Base + Foreground Occlusion composite matches Beauty Master bit-for-bit.

Results: MSE = 0.0000 | RMSE = 0.0000 | PSNR = 100.00 dB (Threshold > 40 dB) | Max Pixel Error = 0.0.

Verdict: Semantic layer decomposition achieved 100% lossless parity without visual degradation.